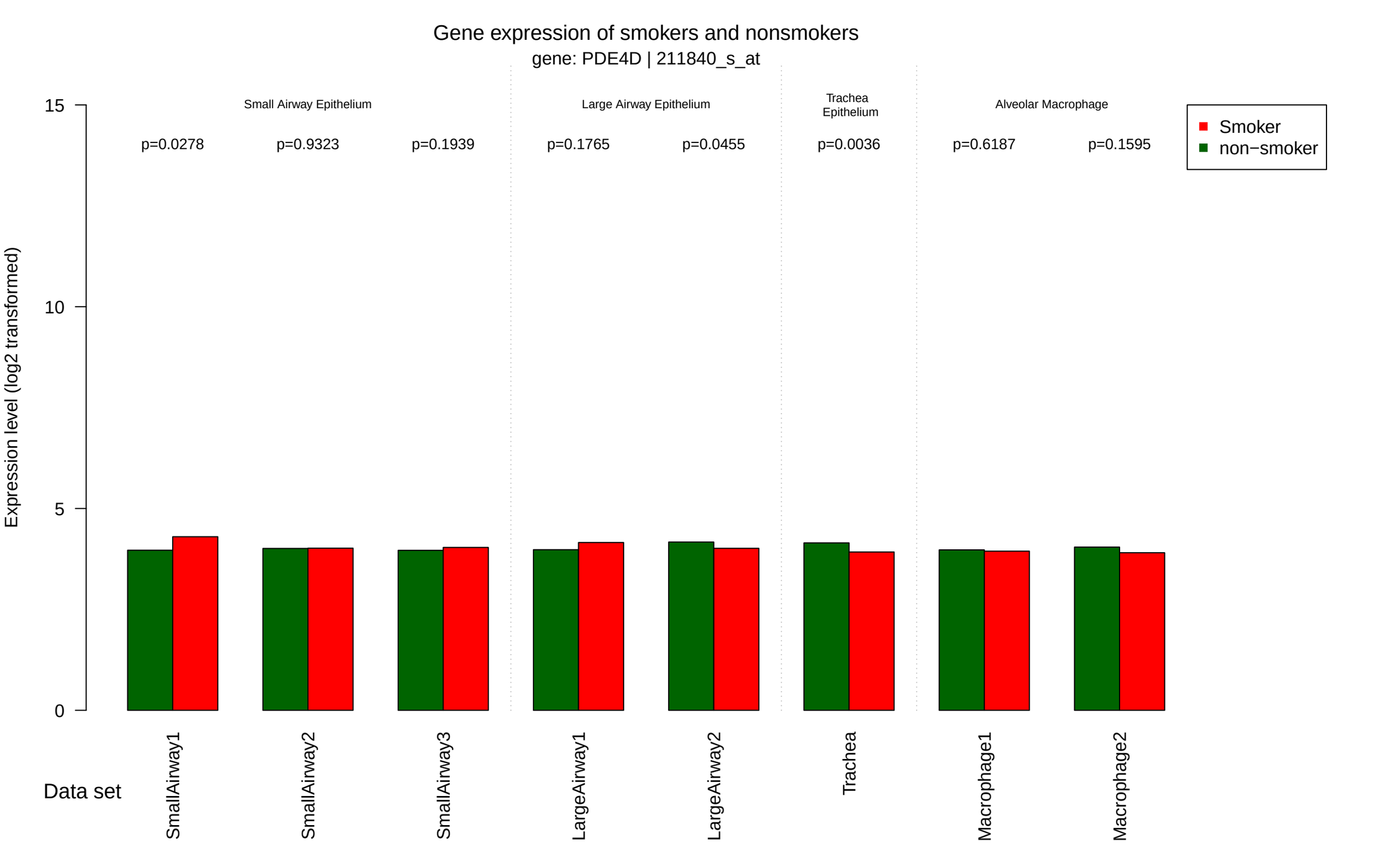
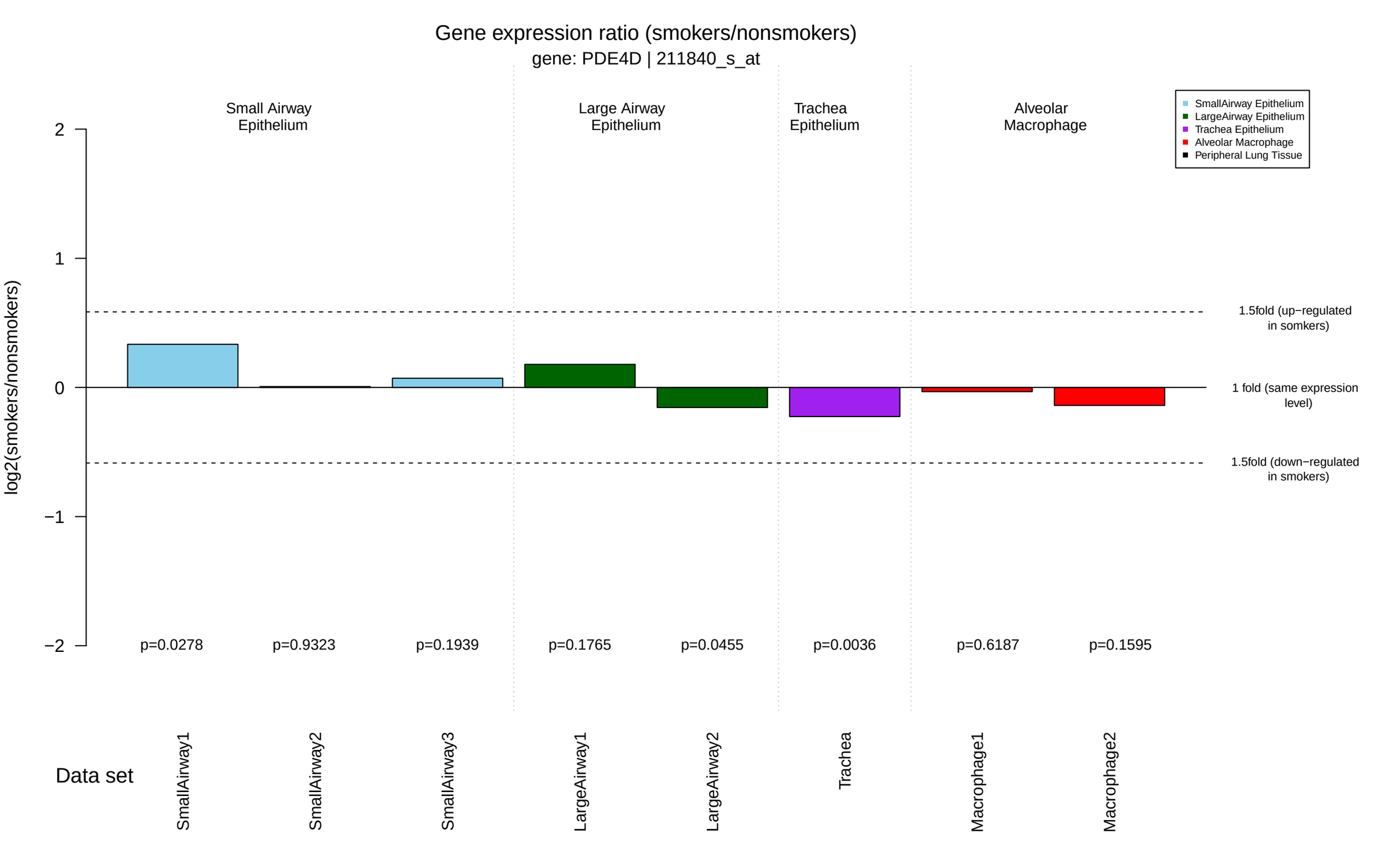
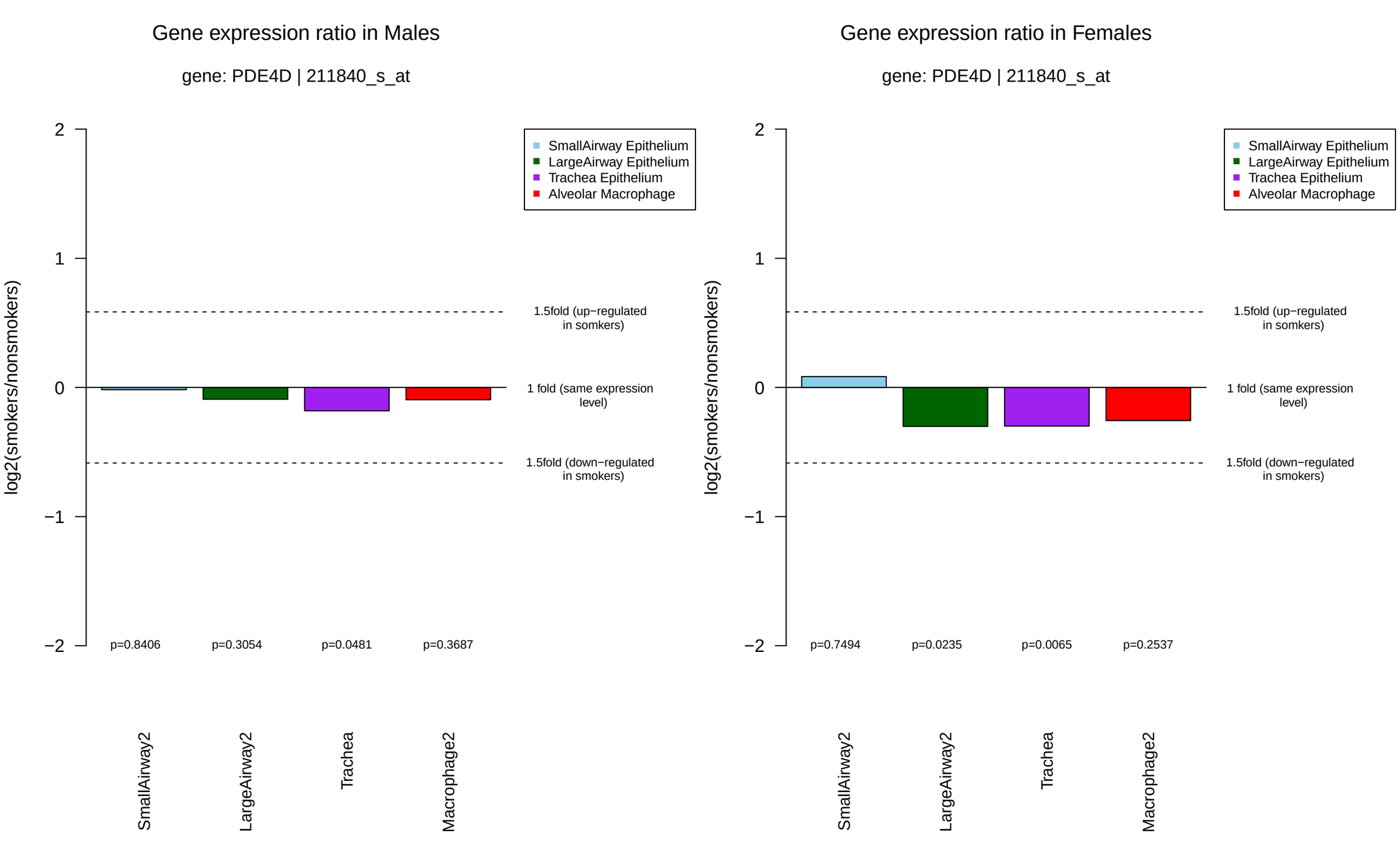
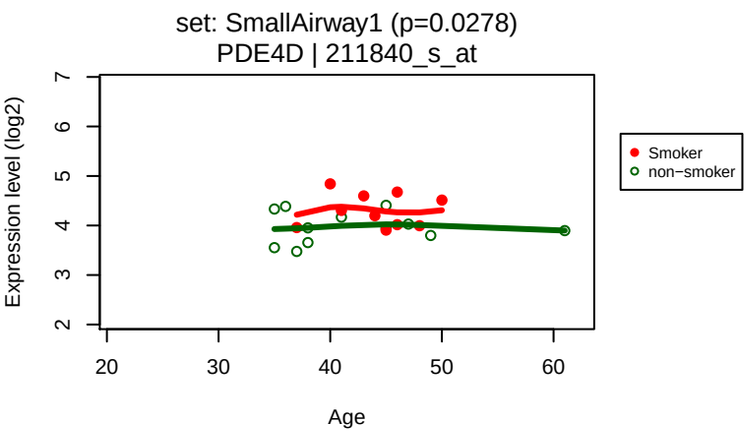
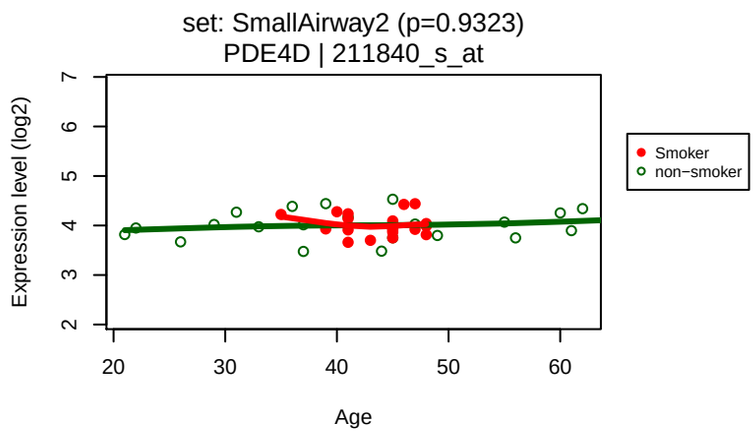
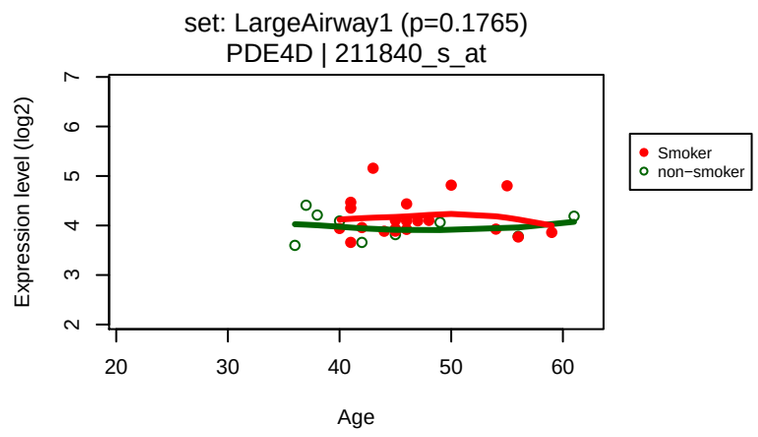

Gene expression of smokers and nonsmokers
gene: PDE4D | 211840_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0278
p=0.9323
p=0.1939
p=0.1765
p=0.0455
p=0.0036
p=0.6187
p=0.1595
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

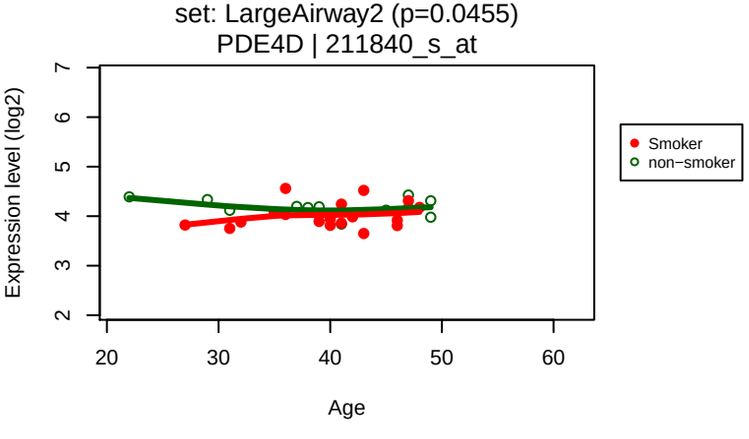

Gene expression ratio (smokers/nonsmokers)
gene: PDE4D | 211840_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0278
p=0.9323
p=0.1939
p=0.1765
p=0.0455
p=0.0036
p=0.6187
p=0.1595
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

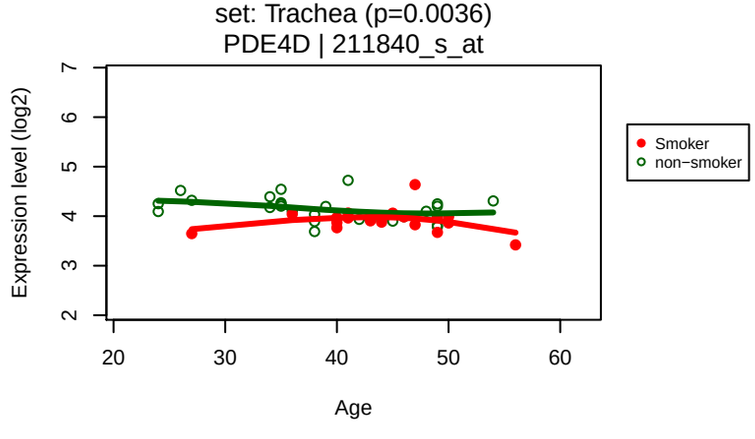

Gene expression ratio in Males
gene: PDE4D | 211840_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8406
p=0.3054
p=0.0481
p=0.3687
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PDE4D | 211840_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7494
p=0.0235
p=0.0065
p=0.2537
SmallAirway2
LargeAirway2
Trachea
Macrophage2

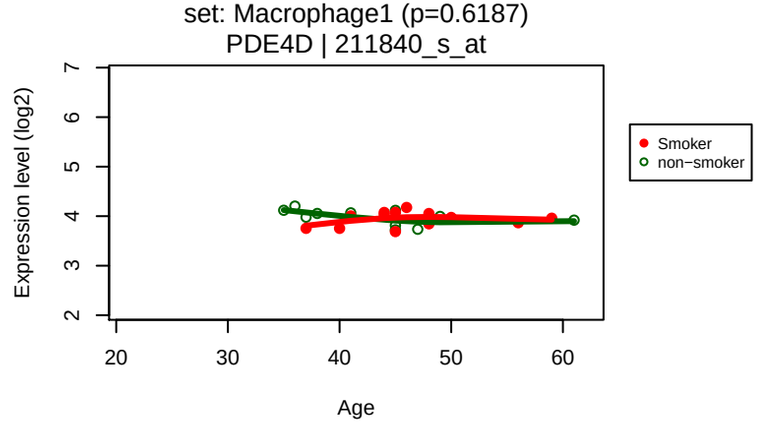

set: SmallAirway1 (p=0.0278)
PDE4D | 211840_s_at
Expression level (log2)
Age
Smoker
non−smoker

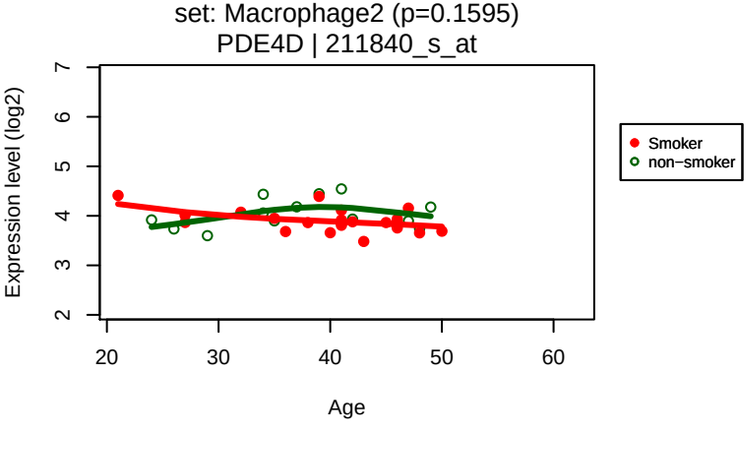

set: SmallAirway2 (p=0.9323)
PDE4D | 211840_s_at
Expression level (log2)
Age
Smoker
non−smoker

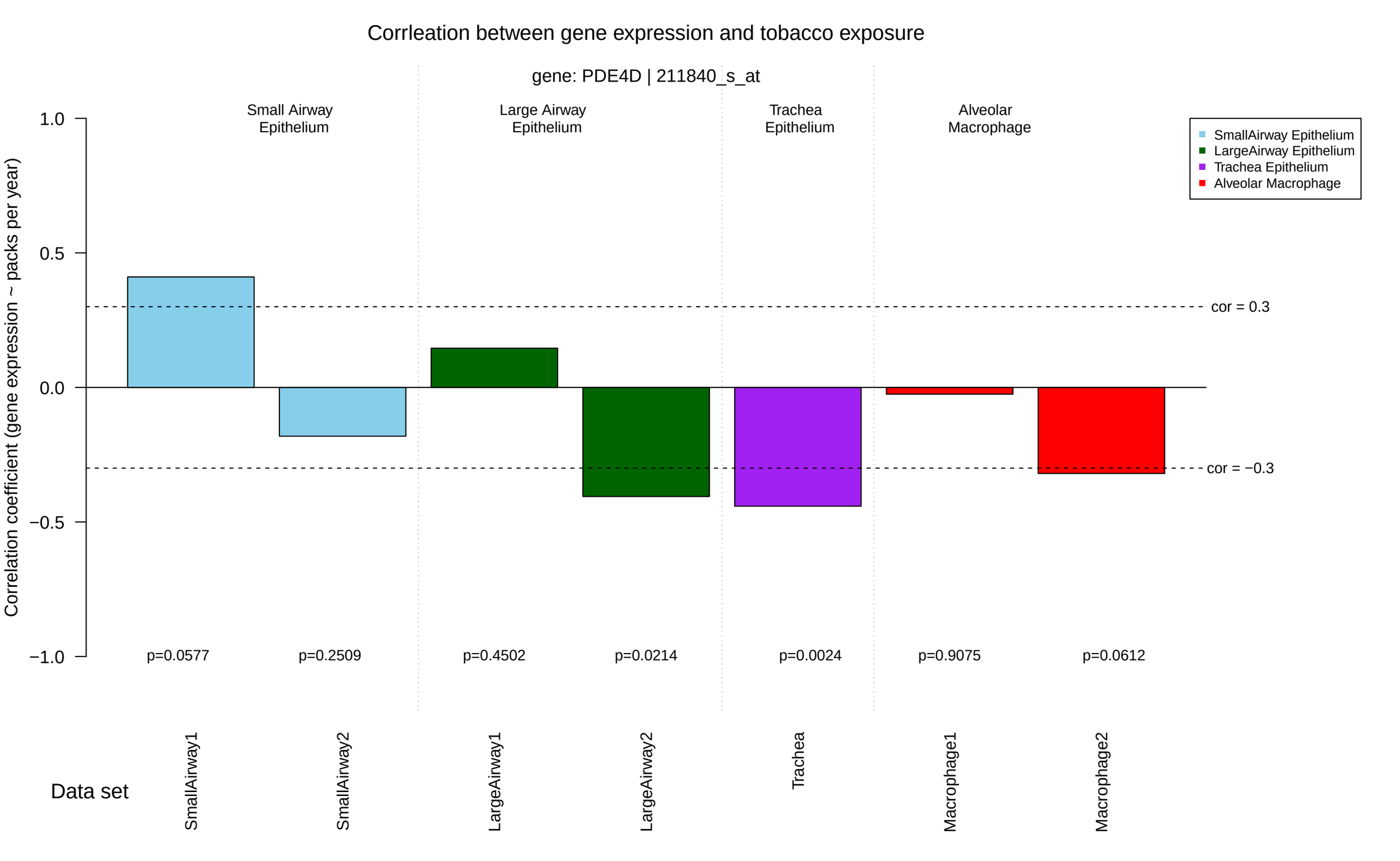

set: LargeAirway1 (p=0.1765)
PDE4D | 211840_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0455)
PDE4D | 211840_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0036)
PDE4D | 211840_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6187)
PDE4D | 211840_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1595)
PDE4D | 211840_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PDE4D | 211840_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0577
p=0.2509
p=0.4502
p=0.0214
p=0.0024
p=0.9075
p=0.0612
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2